Chicken as Reservoir for Extraintestinal Pathogenic Escherichia coli in Humans, Canada

We previously described how retail meat, particularly chicken, might be a reservoir for extraintestinal pathogenic Escherichia coli (ExPEC) causing urinary tract infections (UTIs) in humans. To rule out retail beef and pork as potential reservoirs, we tested 320 additional E. coli isolates from these meats. Isolates from beef and pork were significantly less likely than those from chicken to be genetically related to isolates from humans with UTIs. We then tested whether the reservoir for ExPEC in humans could be food animals themselves by comparing geographically and temporally matched E. coli isolates from 475 humans with UTIs and from cecal contents of 349 slaughtered animals. We found genetic similarities between E. coli from animals in abattoirs, principally chickens, and ExPEC causing UTIs in humans. ExPEC transmission from food animals could be responsible for human infections, and chickens are the most probable reservoir.